miras.commons — Model specification

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Purpose

The model asks one question: to get a resource-saving innovation widely adopted, which group should it start in, and does it matter whether it originates there or is introduced from outside? Groups differ in cultural tightness, parochial altruism, and economic and political power.

Several groups share one common-pool resource, such as water. An innovation reduces each adopter's consumption but costs the adopter something. It can be a product (a water-saving device) or a cultural trait (a convention of using less water). Adoption spreads by social learning within and between groups.

The primary outcomes are how many people end up adopting and whether the resource stays above a collapse threshold. The model is built as `miras.commons` on the miras engine, so the same setup can later be checked for identifiability: which of these mechanisms could a field study actually tell apart?

In scope: one shared resource; a binary trait (adopt or not); groups that differ in size, tightness, parochial altruism and power; learning within and across groups; seeding the innovation in a chosen group; the two modes of origination and introduction.

Not in scope for version 1: several resources or spatial structure within the resource; strategic or forward-looking agents (learners here are myopic imitators); conflict between groups; formal institutions such as elected councils; calibration to a real case. Each could be added later; none is needed to answer the core question.

What the results can and cannot say: the simulation varies tightness, parochialism and power independently, which the real world does not allow, so it can separate their effects. Its answers are conditional on the mechanisms chosen below, and the expected result is a map of which seeding strategy works under which conditions, not a single best group.

Decisions for you

Twelve choices shape what the results mean; each has a proposed default, and the code will keep the alternatives as switches where that is cheap. D1, D2 and D3 matter most: they decide whether adoption can pay privately at all, and whether "originated" and

"introduced" are different experiments.

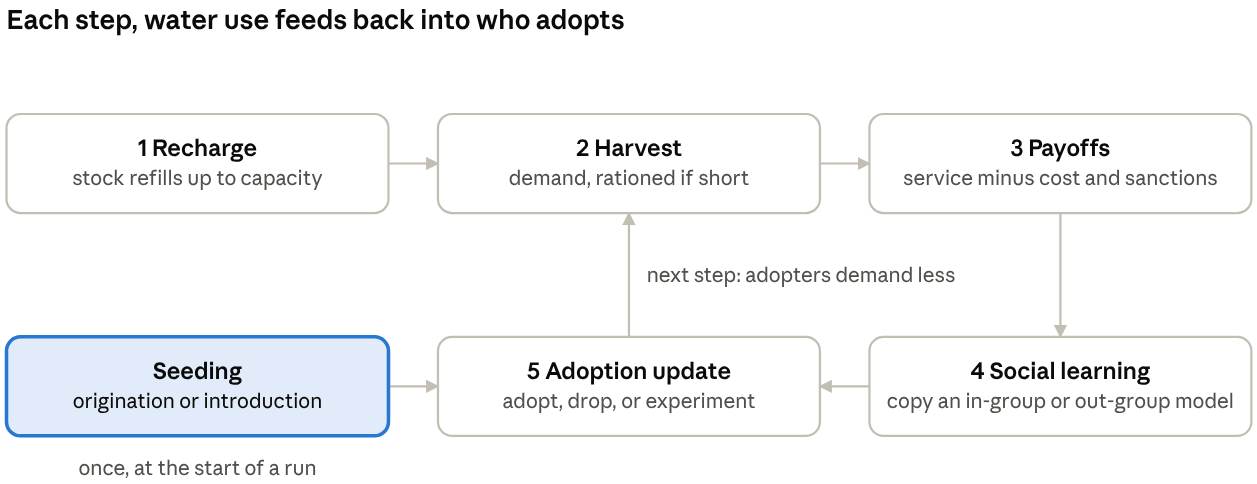
#	Decision	Options	Proposed default	Why it matters
D1	Efficiency or sufficiency	(a) efficiency: same service from less water, like a water-saving device; (b) sufficiency: less water and less service, like a convention of shorter showers	Decided: (a) efficiency for all experiments; (b) kept as a switch for later	Under (a) adoption pays privately when water is rationed; under (b) it never pays privately and survives only through norms and altruism
D2	Rationing under shortage	(a) equal quota per person within a group; (b) cut in proportion to demand; (c) no rationing: first come, first served	(a)	Under (a) with D1(a), scarcity makes adoption privately worthwhile; under (b) it does not
D3	What makes "originated" differ from "introduced"	(a) an ownership tag other groups react to; (b) who the first adopters are; (c) outside support such as a subsidy or persistent outside demonstrators	All three as switches; the main comparison uses (a) + (c)	Without at least one, the two experimental cases are the same starting state
D4	How out-groups react to an owned trait	(a) not at all; (b) copy it less, in proportion to their parochialism; (c) actively reject it as a marker of the other group	(b); (c) as a variant	(c) can make seeding a disliked group backfire

#	Decision	Options	Proposed default	Why it matters
D5	Resource dynamics	(a) constant recharge up to a capacity, like an aquifer or reservoir; (b) logistic regrowth, like a fishery or forest	(a)	(a) gives a clean sustainability threshold: total demand must not exceed recharge
D6	Which tightness mechanisms	conformity to the group majority; sanctions on deviants; less experimentation with new traits	All three, each switchable	Together they make tight groups slow to start and strong to hold, the trade-off to test
D7	How parochial altruism values the commons	in-group members value their own group's share of the water an adopter saves	Yes	Makes parochial altruism favour adoption most in groups with a large share of the resource
D8	What political power does	prestige (others copy them more); a larger share of the resource under shortage; imposing adoption on others	Prestige + larger share; imposition off in version 1	Power can speed diffusion and also let a group avoid scarcity, removing its own reason to adopt
D9	What economic power does	the trait's cost relative to wealth	On for a product, off for a convention	Rich groups find a device cheap; a convention costs everyone the same effort

#	Decision	Options	Proposed default	Why it matters
D10	Learning rule	each learning event is conformist with probability equal to the group's tightness, otherwise payoff-biased imitation of one model	As stated	Ties tightness to how much people follow the majority rather than payoffs
D11	Population dynamics	(a) no births or deaths, a fraction relearns each step; (b) the paper's age structure	(a)	Ages add parameters without adding to this question
D12	Movement between groups	none: ethnic or religious membership is fixed	None	Keeps group identity, and so parochialism, meaningful

Model overview

The model is a loop: adoption lowers demand, demand drains the shared stock, scarcity changes payoffs, and payoffs and norms drive who adopts next. The experiments change only one thing, where and how the innovation is seeded.



one time step · 5 stages and the seeding intervention

Each step runs boxes 1 to 5 in order; seeding happens once, before the first step, and is the only thing the two experimental cases vary.

Entities and state

Entity	State	Changes during a run?
Individual *i*	group *g(i)*; adopter or not, a_i ∈ {0, 1}; payoff this step	adoption: yes; group: no (D12)
Group *g*	size N_g; tightness τ_g; in-group altruism α_i_g; out-group parochialism α_o_g; wealth w_g; political power p_g; contact rate c_g	no: these are the experimental attributes
Resource	stock R_t, capacity K, recharge I per step	stock: yes
Innovation	owner: a group, "outside", or none; cost κ; saving δ; subsidy schedule	owner: no; subsidy: may end

All attributes lie in [0, 1] except sizes, the resource quantities and the cost, so groups can be placed anywhere on each axis independently.

The common-pool resource

The commons is sustainable exactly when total demand does not exceed recharge. With the default recharge model (D5a) that happens once the adoption share reaches A* = (1 − ρ) / δ, where ρ is recharge relative to demand without the innovation.

Demand. Each person needs one unit of service per step. A non-adopter draws d_0 water for it; an adopter draws $d_0(1 - \delta)$, where δ is the innovation's saving.

```
d_i = d_0 (1 - \delta a_i), \qquad D_t = \sum_i d_i
```

Stock. Each step the stock recharges by I up to capacity K, then the harvest H_t is removed. Everyone's demand is met while the stock lasts; otherwise the whole remaining stock is shared out.

```
R_t \leftarrow \min(K, R_{t-1} + I), \qquad H_t = \min(D_t, R_t), \qquad R_t \leftarrow R_t - H_t
```

Scale. Two ratios set the regime: $\rho = I / (N d_0)$, recharge relative to demand with no adopters, and K / I, how many steps of recharge the store holds. A social dilemma needs $\rho < 1$ (without the innovation the stock runs down), and a solution needs $\rho \geqslant 1 - \delta$ (universal adoption would suffice). K / I sets how long the buffer lasts before shortages bite.

```
D_t \le I \iff A_t \ge A^* = \frac{1 - \rho}{\delta}, \qquad A_t = \frac{1}
{N} \sum_i a_i
```

Sharing a shortage between groups. When $H_t < D_t$, group *g* is entitled to a share proportional to $N_g \, p_g^{\gamma}$, where p_g is political power and $\gamma \geqslant 0$ how much power counts ($\gamma = 0$: equal per person across groups). No group gets more than it demands; any excess is passed on to the others in proportion to their entitlements (water-filling).

Sharing within a group (D2a). Each member gets an equal quota q_g, capped at their own demand, again by water-filling: $h_i = \min(d_i, q_g)$.

Service. Under efficiency (D1a) the innovation gives the same service from less water, so a shortage hurts adopters less. Under sufficiency (D1b) adopters accept less service even when water is plentiful.

```
\text{efficiency: } s_i = \min\!\left(1, \frac{h_i}{d_i}\right), \qquad
\text{sufficiency: } s_i = \frac{h_i}{d_0}
```

For example, with a quota $q = 0.6\, d_0$ and $\delta = 0.4$, a non-adopter gets 60% of the service they need and an adopter gets all of it: under efficiency, rationing is what makes the innovation pay privately.

Collapse. The run records a shortage in every step with $H_t < D_t$, and calls the commons collapsed if the mean shortage over the last 100 steps exceeds a threshold (default 10% of demand).

Payoffs

While water is plentiful the innovation is a pure cost to its user, so only norms and in-group altruism can sustain it; once rationing starts, efficiency can make it pay. That is the social dilemma the seeding strategies have to overcome.

Each step person *i* receives a material payoff π_i and acts on a perceived utility u_i, which adds what in-group altruists care about:

$$\pi_i = b\, s_i - \kappa_i a_i - \sigma_i, \qquad u_i = \pi_i + \alpha^{\mathrm{in}}_{g}\, \omega_g\, \lambda(R_t)\, \delta d_0\, a_i$$

Service value b. The value of a full unit of service; b = 1 sets the scale of every other payoff.

Cost κ_i. For a product (D9 on) the cost falls with the group's wealth; for a convention it is the same effort for everyone (ε = 0). Under introduction with a subsidy, the cost is reduced by a fraction φ until the subsidy ends at step T_sub.

$$\kappa_i = \frac{\kappa\, (1 - \varphi\, [t < T_{\mathrm{sub}}])}{w_{g}^{\varepsilon}}$$

Sanction σ_i (tightness). A person whose choice differs from their group's majority pays a sanction that grows with the group's tightness τ_g and with how settled the norm is (f_maj = the majority's share). A new trait in a tight group is therefore punished while rare, and protected once it becomes the majority.

$$\sigma_i = \tau_g\, \sigma_{\max}\, (2 f_{\mathrm{maj},g} - 1)\, [a_i \ne \text{majority}_g]$$

In-group altruism. Water an adopter saves (δ d_0 per step) stays in the common stock. Group *g* will get a share ω_g of it, its entitlement share N_g p_g^γ / Σ_h N_h p_h^γ. An in-group altruist (α^{in}_g > 0) values that part at λ(R), a price of water that rises as the stock falls: $\lambda(R) = \lambda_0 (1 - R / K)$. Water saved for other groups is not valued: that is what makes the altruism parochial.

Two consequences follow before any simulation. Parochial altruism favours adoption more in groups with a large entitlement share, which usually means large or powerful groups. And that pull strengthens as the stock falls.

What each quantity is used for. Learning compares perceived utilities u (people act on what they value). Reported wealth and inequality use material payoffs π.

Group attributes as mechanisms

Each attribute acts through two or three named mechanisms, each with its own switch, so every result can be traced to the mechanism that produced it. No attribute is given a direct effect on adoption.

Attribute	Mechanism	Where it acts	Switch
Tightness τ_g	conformity: learning events follow the group majority with probability τ_g, at exponent θ_{max}	learning (D10)	`tight_conformity`
	sanctions on deviating from the group majority	payoff σ_i	`tight_sanctions`
	less experimentation: spontaneous trial of the other option at rate $\mu(1 - \tau_g \eta)$	adoption update	`tight_caution`
In-group altruism α^{in}_g	values own group's share of saved water	perceived utility u_i	`ingroup_value`
Out-group parochialism α^{out}_g	accepts an out-group model only with probability $1 - \alpha^{out}_g$	learning	`outgroup_filter`
	resists a trait owned by another group or by outsiders (D4)	learning	`ownership_bias`
Wealth w_g	lowers the effective cost of a product	payoff κ_i	`wealth_cost`
Political power p_g	prestige: out-group learners pick this group's members as models in proportion to p_g^{β}	learning	`power_prestige`
	a larger share of the water in a shortage, via γ	resource allocation	`power_share`
Contact c_g	how often members learn from other groups at all	learning	always on

Built-in tensions, to be tested rather than assumed. Tightness should slow a trait's start (sanctions and caution against a rare option) and protect it once it is the norm. Power should speed diffusion through prestige, yet a powerful group that is shielded from shortage ($\gamma > 0$) has less private reason to adopt. Parochialism should help adoption inside

the seeded group and block its spread beyond. If a result depends on one of these tensions, switching that mechanism off should remove it; the analysis will report those checks.

Contact and tightness. Tight cultures are often less open to outsiders. In version 1 contact is a separate attribute so the two can be varied apart; a variant can tie them.

Social learning

The innovation crosses group boundaries only through learners who pick an out-group model, so contact, prestige and parochialism together set its route between groups. Within a group, tightness decides whether people follow the majority or compare payoffs.

Each step, a fraction p_learn of every group are learners. For each learner *i* in group *g*:

1. **Experiment.** With probability $\mu_g = \mu\,(1 - \eta\,\tau_g)$ the learner tries the other option and skips the rest. Tight groups experiment less.
2. **Conformist or payoff-based.** With probability τ_g the event is conformist: the learner samples n in-group members and adopts with probability $f^\theta / (f^\theta + (1-f)^\theta)$, where f is the adopting share of the sample and $\theta = \theta_{max}$. This is miras's existing `Conformity` rule for two options. Otherwise go to step 3.
3. **Pick one model.** With probability c_g the learner looks outside the group: another group *h* is chosen with probability proportional to $N_h\, p_h^\beta$ (prestige), and a random member of *h* becomes the model, kept with probability $1 - \alpha^{out}_g$. If the out-group model is rejected, or with probability $1 - c_g$, the model is a random in-group member.
4. **Compare and switch.** If the model's option differs, the learner switches with probability $\mathrm{clip}(\beta_s\,(u_{model} - u_i), 0, 1)$. This is proportional imitation, as in miras's `PayoffBias`, now with utilities that include water, cost, sanctions and in-group valuation.
5. **Ownership bias (D4).** When the switch would adopt an innovation owned by another group, its probability is multiplied by $1 - \zeta\,\alpha^{out}_g$; for an innovation owned by "outside", by $1 - \zeta_{ext}\,\alpha^{out}_g$, and this applies in every group, the seeded one included. Under D4(c), adopters of an out-group-owned trait also lose $\zeta\,\alpha^{out}_g$ in utility each step, so the trait becomes a costly marker of the other group.

All learners update together at the end of the step.

A known confound, kept on purpose. A learner copying a model from a powerful group compares utilities, and that model's utility is higher partly because its group is shielded from shortage. Learners can therefore copy a trait because of who holds it rather than

what it does. This is success-biased learning as it happens in real populations; the analysis can separate it by turning `power_share` off.

Origination versus introduction

Both cases start with the same adopters in the same group; they differ only in who is seen to own the innovation and whether outside support exists. Every difference in outcome between the two cases is therefore caused by those two features.

At step 0, k adopters are placed in the seed group g* (default k = 5% of N_g*, chosen at random), and nobody else adopts.

Feature	Origination in g*	Introduction into g*
Owner of the innovation	group g*	"outside"
Resisted by g* itself	no	yes, in proportion to α^{out}_g* and ζ_ext
Resisted by other groups	yes, in proportion to their α^{out} and ζ	yes, in proportion to their α^{out} and ζ_ext
Subsidy on the cost	none	fraction φ off, for groups that adopt, until step T_sub
Outside demonstrators	none	an outside pool of adopters that g* learners can pick as models (weight p_ext) until step T_demo
First adopters	random members of g*	random members of g* (D3b adds targeted options)

Two reference runs frame every comparison: no seeding at all (the commons without the innovation), and seeding every group equally (the best a simultaneous campaign could do).

What the contrast measures. With ζ = ζ_ext and no outside support, the two cases differ only in whether the seed group itself resists the trait. Turning on the subsidy and the demonstrators adds what outside introducers usually bring. Reporting both separates "being foreign" from "coming with help".

Targeted first adopters (D3b), optional. Choosing elites or well-connected members first needs individual differences within groups, which version 1 does not have. The option adds a within-group status score used as prestige in model choice; it is off by default.

Outcomes and experimental design

The main experiment varies the seed group's five attributes on a full factorial, crossed with the two seeding modes and four environmental conditions: 1,944 cells, 50 runs each. Its answer is a map of the best seeding strategy for each condition.

Outcomes, recorded every run

Outcome	Definition	Answers
Final adoption A_T	adopting share of everyone at the last step T	how many people use it
Sustained	mean adoption over the last 100 steps $\geq A^*$ and not collapsed	whether the commons is governed sustainably
Collapse	mean shortage over the last 100 steps > 10% of demand	whether the commons failed
Reach	number of groups where more than half adopt	whether it spread beyond the seed group
Time to A^*	first step with $A_t \geq A^*$	how fast
Inequality	Gini coefficient of mean material payoff across groups	who bears the shortage
Adoption by group	final adoption share in each group	where it took hold

E1: which seed group works best. Five groups of 200 people. Four background groups sit at the middle level of every attribute; the seed group's tightness, in-group altruism, out-group parochialism, wealth and power each take three levels (low, middle, high): $3^5 = 243$ seed profiles. Each is run in both modes (origination, introduction with support) under 2 × 2 conditions: water stress ρ (mild or severe) and contact (low or high), with the innovation always an efficiency (D1). That gives 243 × 2 × 4 = 1,944 cells × 50 replicates = 97,200 runs of 500 steps.

E1b: does the background matter? Seeding a tight group among loose groups may differ from seeding it among tight ones. A smaller run repeats the most informative E1 cells with the background groups set low and high.

E2: real configurations. In real settings the attributes are correlated (for example, a powerful group that is also loose). You supply group profiles for a case you have in mind, and the experiment compares seeding each actual group in each mode.

E3: mechanism knockouts. For every clear E1 result, rerun with each relevant mechanism switched off (see the switches under Group attributes) to find the mechanism responsible.

Analysis. For each outcome: the mean and a 95% interval per cell over replicates; main effects and two-way interactions of the seed attributes, per condition; and the contrast between modes, introduction minus origination, per seed profile. The headline figure is a map: for each condition, which seed profile and mode maximise sustained adoption.

Cost. One run of 1,000 agents for 500 steps should take well under a second, so E1 is roughly 5 to 15 CPU-hours: a few hours on your 8-core Mac. It uses the checkpointing from 0.1.1.

Verification

No experiment runs until the model reproduces nine results that are known without it, each an automated test. This is the same standard the miras engine already meets against population-genetic theory.

#	Setup	Known result
V1	No adopters, learning off, full stock	after the first step the stock falls by $N d_0 - I$ per step, emptying after about $K / (N d_0 - I)$ steps when $\rho < 1$; it never empties when $\rho \geq 1$
V2	Everyone adopts, learning off	sustained if and only if $\rho \geq 1 - \delta$
V3	Any shortage	water is conserved ($\Sigma h_i = H_t$), nobody gets more than they demand, groups get entitlements in proportion to N_g p_g^γ, and γ = 0 gives equal quotas across groups

#	Setup	Known result
V4	One group, plenty of water, no norms or altruism	adoption declines as the replicator equation with payoff difference $-\kappa$ predicts (the same test `PayoffBias` passes now)
V5	One group, rationing, efficiency, no norms	adoption rises while a non-adopter's service shortfall $b\,(1 - q / d_0)$ exceeds κ, and stops where they balance
V6	Conformity only ($\tau = 1$, sanctions and caution off)	a seeded minority below half disappears, a majority above half goes to fixation
V7	Identical groups	the seed group makes no difference: outcome distributions match across seed choices
V8	Out-group parochialism 1, no experimentation, no demonstrators	the trait never leaves the seed group
V9	Any run	same seed gives identical results, and an interrupted run resumes to identical results

V1 to V3 test the resource on its own and V4 to V6 the learning, both against analytic answers; V7 to V9 are symmetry and bookkeeping checks.

Identifiability follow-up

If the experiments say, for example, that seeding the tight group works best, a field study would have to show that tightness, and not power or parochialism, drives adoption in a real case. miras's `identify` checks, before any data are collected, whether a proposed study could tell those mechanisms apart.

The model's attributes become the parameters to infer, and a study design becomes the set of summaries it would measure. `identify` then reports `EQUIFINALITY` where two mechanisms produce the same data (for example, a high sanction strength and a high conformity weight both yield sharp, stable majorities) and `NOT_IDENTIFIED` where a design carries no information about a parameter.

Candidate designs to compare, from cheapest to richest:

1. **One survey:** adoption share per group at one time.
2. **Repeated surveys:** adoption per group over several years, which shows the speed and route of spread.
3. **Surveys with network questions:** who learned from whom, including across groups, as in the paper's longitudinal example.
4. **Adding sanction and norm measures:** whether non-adopters report disapproval, which bears directly on tightness.

The expected product is a recommendation of the cheapest design that separates the mechanisms the simulation found to matter. This step comes after E1 to E3, because only then is it known which mechanisms need separating.

Parameters

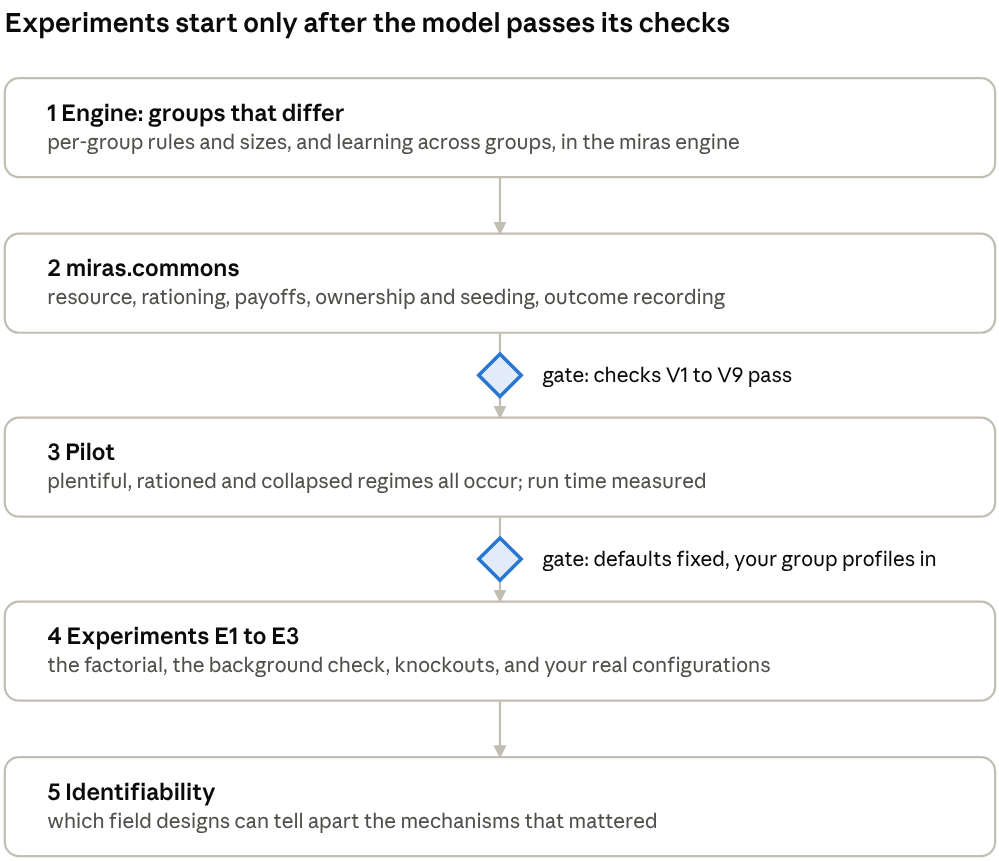
The defaults below are provisional: a pilot run will check that each regime (plentiful, rationed, collapsed) actually occurs within the ranges, and they will be adjusted before E1.

Symbol	Meaning	Default	Range or levels
G, N_g	number of groups, people per group	5, 200	N_g may differ in E2
T	steps per run	500	
k	first adopters in the seed group	5% of N_g*	1% to 20%
d_0, δ	demand without the innovation, fraction it saves	1, 0.4	δ 0.2 to 0.6
ρ	recharge relative to demand without the innovation	0.85 (mild), 0.7 (severe)	must lie between 1 − δ and 1
K / I	steps of recharge the store holds	20	5 to 50
b	value of one unit of service	1	fixed (sets the scale)
κ	cost of the innovation per step	0.1	0 to 0.3
ε	how cost scales with wealth	1 (product), 0 (convention)	0 or 1

Symbol	Meaning	Default	Range or levels
σ_max	largest sanction for deviating	0.3	0 to 0.6
λ_0	value of saved water at an empty store	1	0 to 3
p_learn, n	fraction learning per step, conformist sample size	0.1, 10	
θ_max	conformity exponent in conformist events	3	1 to 5
β_s	slope of payoff-based switching	1	0.5 to 2
μ, η	experimentation rate, how much tightness suppresses it	0.01, 0.8	
β, γ	weight of power in prestige, in the shortage share	1, 1	0 to 2
c_g	contact with other groups	0.1 (low), 0.3 (high)	0 to 0.5
τ_g, α^{in}_g, α^{out}_g	tightness, in-group altruism, out-group parochialism	0.5	0.1, 0.5, 0.9
w_g, p_g	wealth, political power (relative)	1	0.5, 1, 2
ζ, ζ_ext	resistance to a trait owned by another group, by outsiders	0.5, 0.5	0 to 1
φ, T_sub	subsidy fraction and duration	0.5, 100 steps	
p_ext, T_demo	prestige and duration of outside demonstrators	2, 100 steps	
collapse threshold	mean shortage over the last 100 steps	10% of demand	

Implementation plan

The work ships as miras 0.2.0 in five phases, and nothing is run for results until the model passes its verification checks. Two inputs are needed from you: answers to the decisions above before phase 1, and any real group profiles for E2 before phase 4.



implementation plan · 5 phases, 2 gates

Phase 1 adds general features to the engine, so they serve other models too: per-group learning rules and parameters, unequal group sizes, and cross-group model choice weighted by prestige and filtered by parochialism. Phase 2 builds everything specific to this problem in a new module, `miras.commons`. Both come with tests in the style of the current suite, and the release follows the existing TESTING and PUBLISHING procedure.

References

The works the mechanisms draw on, cited from memory: check volume and page details before citing them in a paper.

- Boyd, R. & Richerson, P. J. (1985). *Culture and the Evolutionary Process.* University of Chicago Press. Conformist and payoff-biased transmission.
- Choi, J.-K. & Bowles, S. (2007). The coevolution of parochial altruism and war. *Science* 318, 636–640. Parochial altruism as in-group altruism combined with out-group hostility.

- Deffner, D., Fedorova, N., Andrews, J. & McElreath, R. (2024). Bridging theory and data: A computational workflow for cultural evolution. *PNAS* 121(48), e2322887121. The workflow and model miras is built on.
- Gelfand, M. J. et al. (2011). Differences between tight and loose cultures: A 33-nation study. *Science* 332, 1100–1104. Tightness as strong norms and low tolerance of deviance.
- Henrich, J. & Gil-White, F. J. (2001). The evolution of prestige. *Evolution and Human Behavior* 22, 165–196. Prestige-biased copying.
- Ostrom, E. (1990). *Governing the Commons.* Cambridge University Press. Common-pool resources and the conditions for governing them sustainably.
- Schlag, K. H. (1998). Why imitate, and if so, how? *Journal of Economic Theory* 78, 130–156. Proportional imitation, the switching rule used here.